21-242 Recitation 12

Rank–nullity — Q2. Rank–nullity

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Rank-nullity. Prove that if $T : V \to W$ and $\dim(V) < \infty$, $\dim(\mathrm{im}(T)) + \dim(\ker(T)) = \dim(V)$.

Course-level solution

Let $n = \dim V$ and let $(\vec{v}_1, \ldots, \vec{v}_k)$ be a basis of $\ker T$ (possibly $k = 0$). Extend to a basis $(\vec{v}_1, \ldots, \vec{v}_k, \vec{u}_1, \ldots, \vec{u}_{n-k})$ of V (extension lemma).
We claim $\big(T(\vec{u}_1), \ldots, T(\vec{u}_{n-k})\big)$ is a basis of $\mathrm{im}\, T$.
Spanning. For $\vec{v} = \sum a_i \vec{v}_i + \sum b_j \vec{u}_j$,

$$T(\vec{v}) = \sum b_j T(\vec{u}_j)$$

because $T(\vec{v}_i) = \vec{0}$. So the $T(\vec{u}_j)$ span $\mathrm{im}\, T$.
Independence. If $\sum b_j T(\vec{u}_j) = \vec{0}$, then $T\big(\sum b_j \vec{u}_j\big) = \vec{0}$, so $\sum b_j \vec{u}_j \in \ker T = \mathrm{Span}(\vec{v}_1, \ldots, \vec{v}_k)$. Then $\sum b_j \vec{u}_j - \sum c_i \vec{v}_i = \vec{0}$ for some c_i, and independence of the full basis of V forces every $b_j = 0$.
Therefore $\dim \mathrm{im}\, T = n - k = \dim V - \dim \ker T$.

Later-course solution (using ideas from the rest of the handout)

For T_A, this is "number of columns = number of pivot columns + number of free columns", which is visible from RREF. The abstract proof above is the same counting after choosing coordinates.